

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination 2025-26 (NEP)

DEPARTMENT OF GEOLOGY**Paper Code: ESMJ21 / ESMN21 — Mineral and Earth Material**

Time: 2:30 Hours

Maximum Marks: 70

Ref Code: Conf/Even/N/2025-26/51

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Answer five questions in all, selecting two questions each from Section-A and Section-B. Section-C is compulsory. Figures in the right-hand margin indicate marks.***SECTION – A****Question 1**

[14 Marks]

With the help of a neat and well-labelled diagram, describe the components of a polarizing (petrological) microscope and explain the function of each component.

Question 2

[14 Marks]

Answer any **two** of the following ($2 \times 7 = 14$ Marks):

- (a) Briefly describe the method of rock thin-section preparation.
- (b) Discuss the classification of silicate group of minerals.
- (c) Differentiate between the physical properties of feldspar and calcite.

Question 3

[14 Marks]

Describe the different mineral-forming processes. Discuss each process with suitable examples of minerals formed under them.

Question 4

[14 Marks]

Answer the following questions with suitable examples ($4 \times 3.5 = 14$ Marks):

- (a) Why do diamond and graphite have the same chemical composition but different specific gravities?
- (b) Explain why a mineral with cleavage can also show fracture.
- (c) Why is diamond not just ten times harder than talc on the Mohs hardness scale?
- (d) Explain how colour can be a diagnostic property in the identification of certain minerals.

SECTION – B

Question 5

[14 Marks]

Hornblende and Orthopyroxene both possess prismatic crystal faces. Do they belong to the same crystallographic system? Justify your answer. Differentiate between these two minerals based on their optical properties. Is there any difference between a prismatic face and a pyramidal face?

Question 6

[14 Marks]

Explain in brief with suitable examples ($2 \times 7 = 14$ Marks):

- (a) Is it always necessary that parallel crystal faces would have similar Miller indices?
- (b) Is it possible to deduce Weiss parameters from Miller indices?

Question 7

[14 Marks]

How would you distinguish between optically isotropic and optically anisotropic minerals? Provide at least one example each of optically isotropic and anisotropic minerals and also mention their crystal system. Two Olivine crystals (a common gemstone: 'Peridot') excavated from the basaltic rocks of two different depths from an igneous complex – do they possess the same interfacial angles? Justify your answer with suitable reasons.

SECTION – C (Compulsory)

Question 8

[14 Marks]

Define / Answer the following in 1 word or sentence each:

- (a) Name a mineral which has different colour and streak. [1 Mark]
- (b) Name a tectosilicate mineral which contains cleavage. [1 Mark]
- (c) Name a mineral that commonly exhibits third-order interference colours in a standard thin section. [1 Mark]
- (d) Which mineral shows bladed habit? [1 Mark]
- (e) Name a mineral that commonly shows polysynthetic twinning. [1 Mark]
- (f) Define the following ($9 \times 1 = 9$ Marks):
 - (i) Mineral
 - (ii) Plane of symmetry
 - (iii) Axis of rotation
 - (iv) Solid angle
 - (v) Euler's formula
 - (vi) Birefringence
 - (vii) Extraordinary ray
 - (viii) Nicol prism
 - (ix) Silicate polymerization